

Chromatic quasisymmetric functions for signed graphs

Raquel Melgar (LaBRI, University of Bordeaux)

Joint work with Jean-Christophe Aval

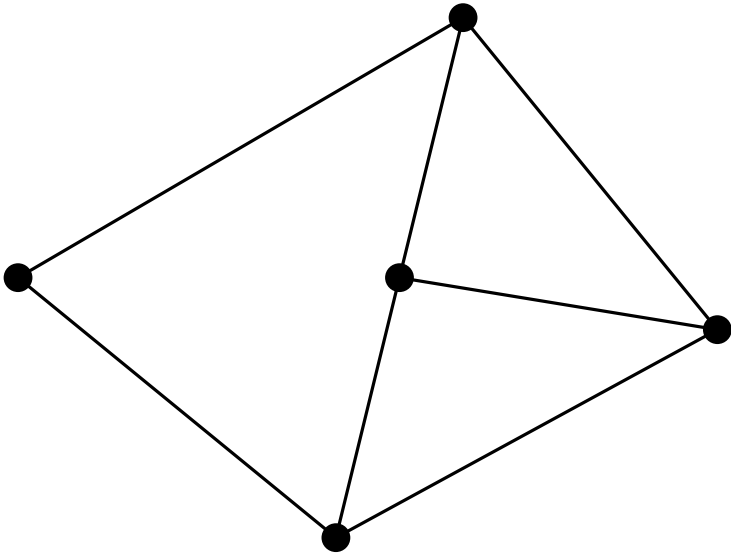
XXI Spanish Meeting in Computational Geometry

Santander, June 30 2025

I.Introduction and preliminaries

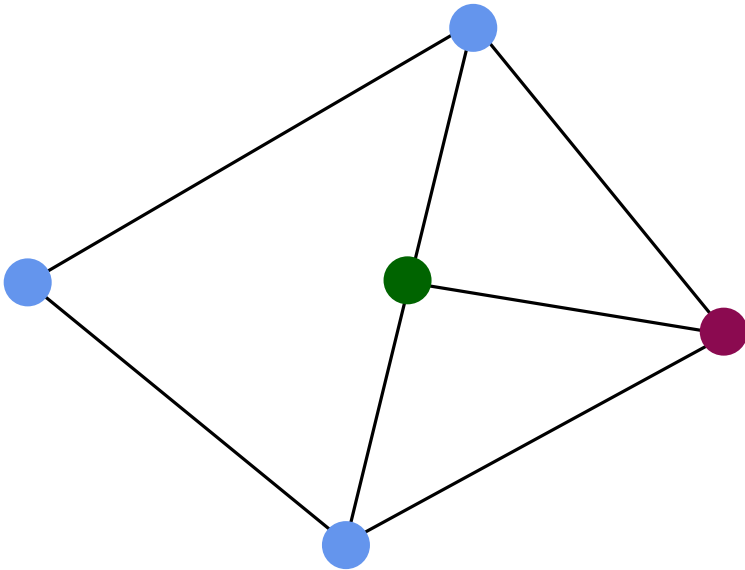
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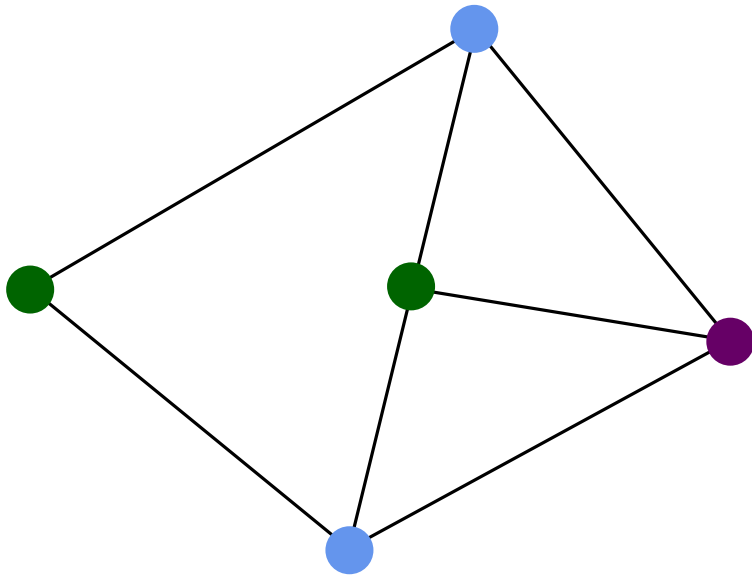
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This is **not** a proper coloring of G

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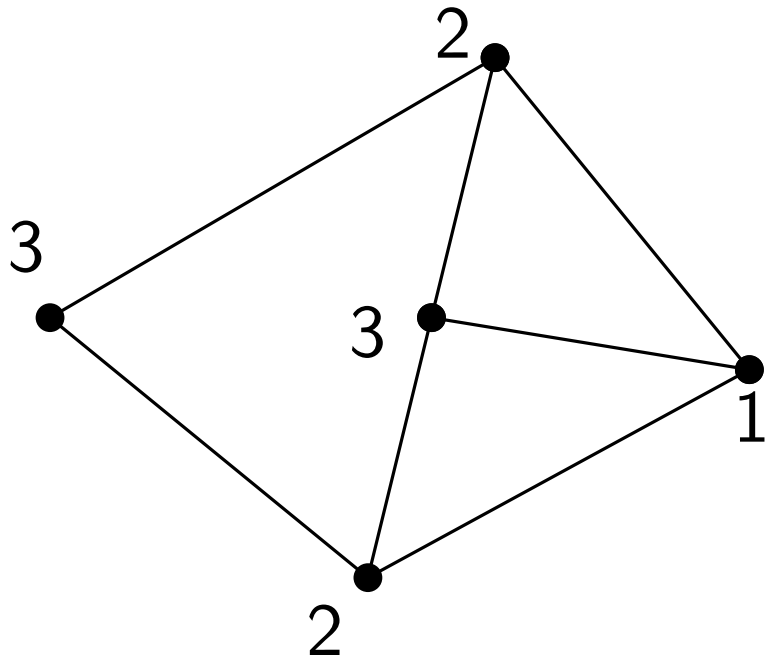
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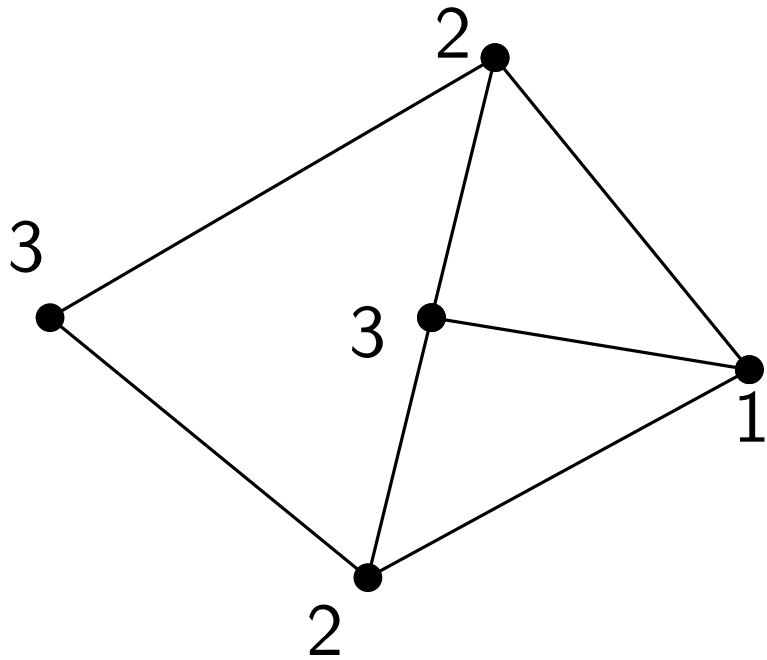
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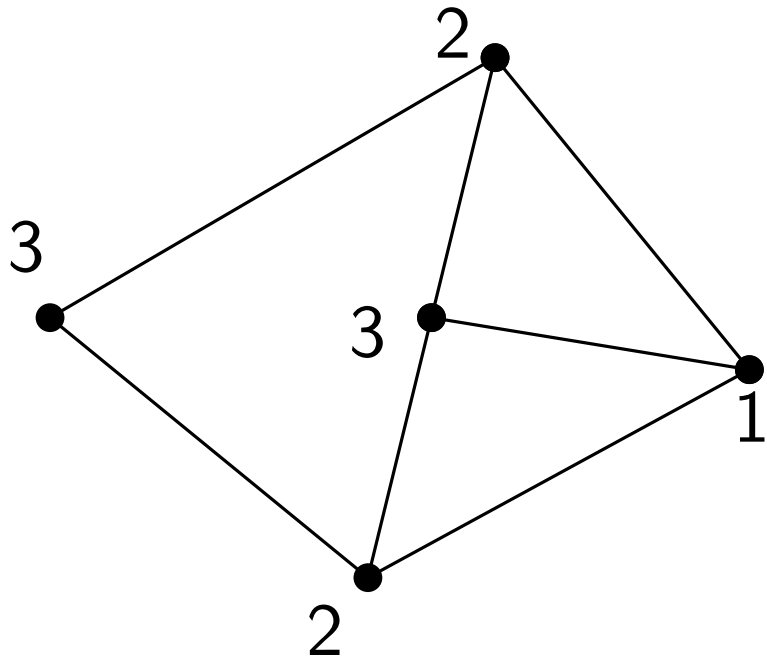


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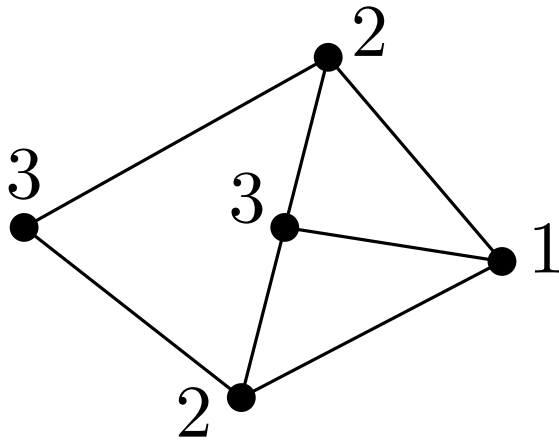
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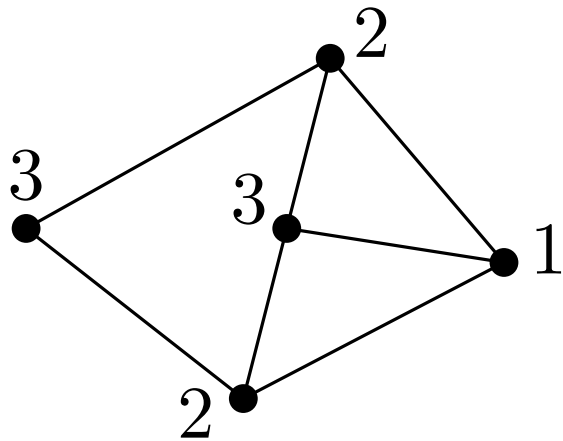
- We will denote χ the *chromatic number* of G .
- $\chi_G(k) :=$ number of (proper) colorings of G with k colors

The chromatic symmetric function

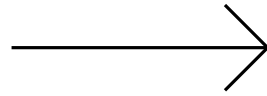


coloring κ

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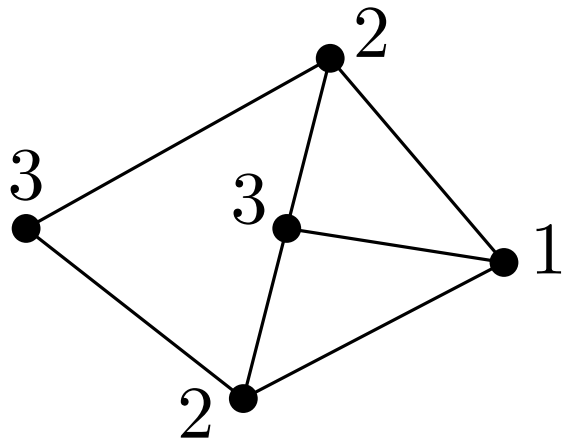


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$$\begin{aligned}x^\kappa &= x_{\kappa(v_1)} x_{\kappa(v_2)} \cdots x_{\kappa(v_n)} \\ &= x_1 x_2^2 x_3^2\end{aligned}$$

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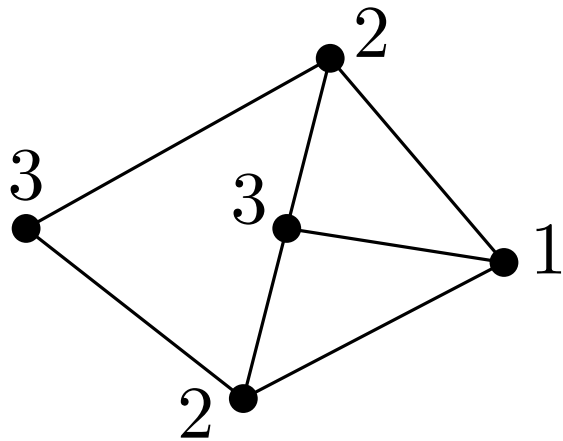
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In 1995 Stanley defined the *chromatic symmetric function* of a graph G as:

$$X_G(x_1, x_2, \dots) = \sum_{\kappa \text{ proper}} x^\kappa \in \text{Sym}^{|V|}$$

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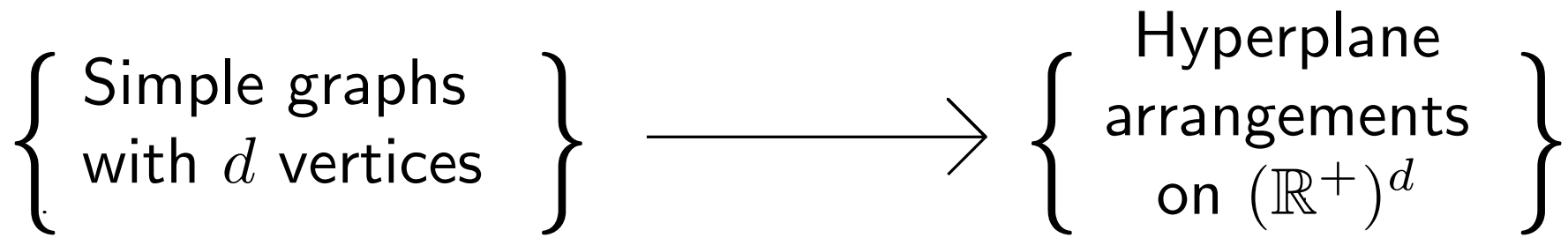
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$$X_G(1, 1, \dots, 1, 0, \dots) = \chi_G(k)$$

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The geometric viewpoint



The geometric viewpoint

$$\left\{ \begin{array}{l} \text{Simple graphs} \\ \text{with } d \text{ vertices} \end{array} \right\} \longrightarrow \left\{ \begin{array}{l} \text{Hyperplane} \\ \text{arrangements} \\ \text{on } (\mathbb{R}^+)^d \end{array} \right\}$$

$$G = (V, E) \longmapsto \mathcal{H}_G = \{x_i = x_j : \{i, j\} \in E\}$$

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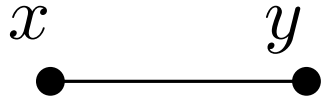
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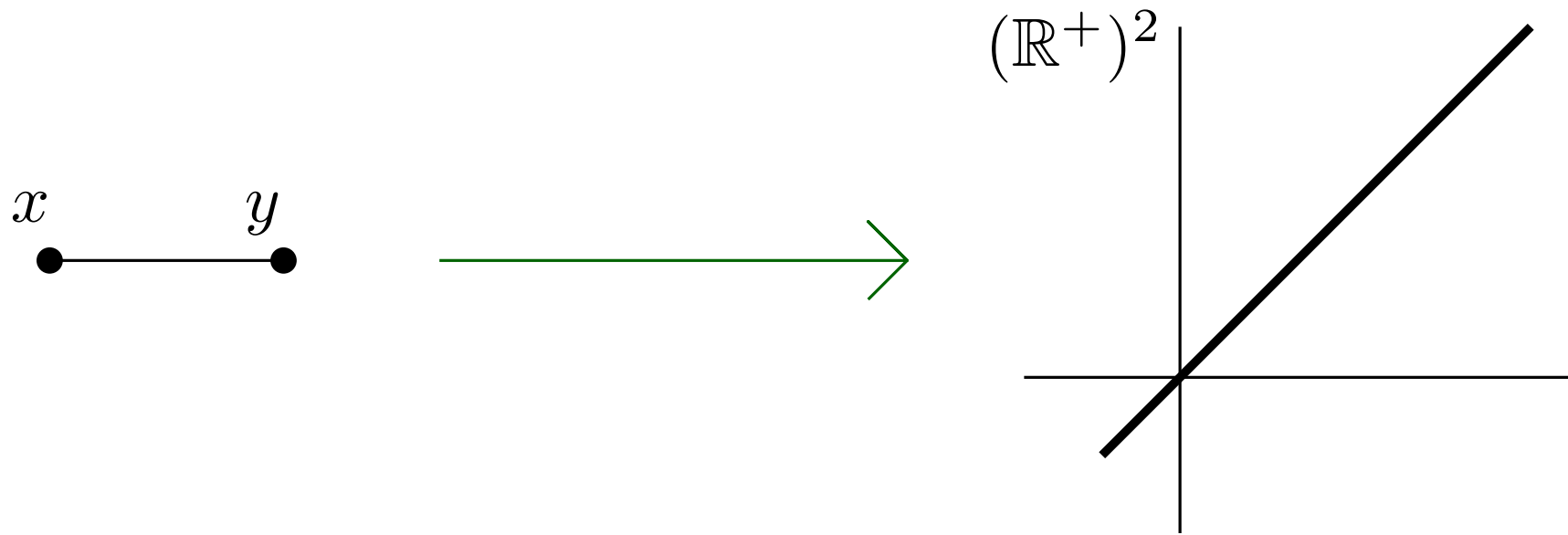
$$\chi_G(k) = \chi_{\mathcal{H}_G}(k)$$

where $\chi_{\mathcal{H}_G}(k)$ is the
characteristic
polynomial of $\mathcal{H}_G$

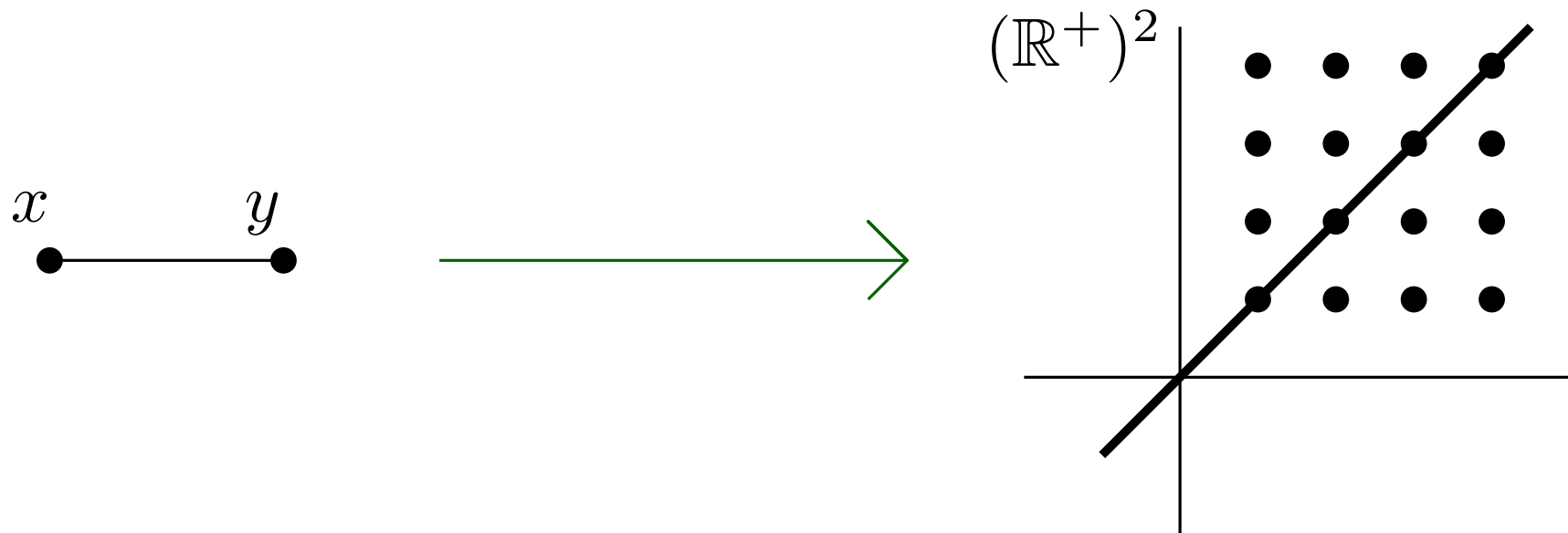
The geometric viewpoint: an example



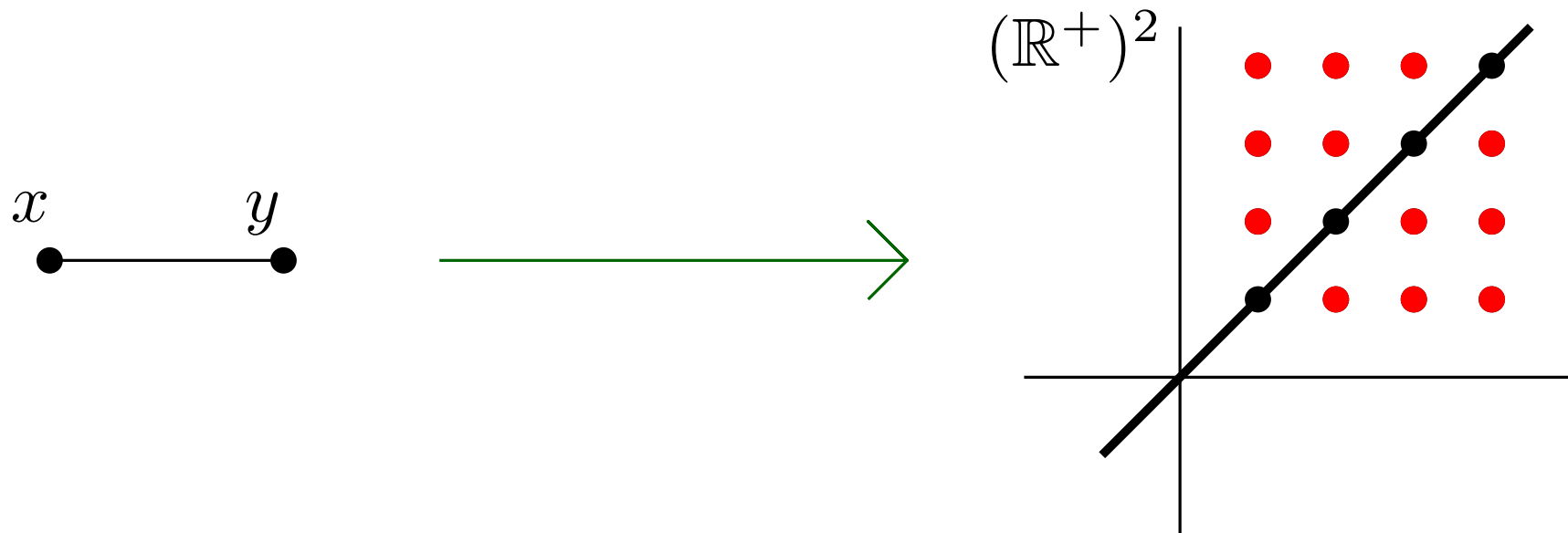
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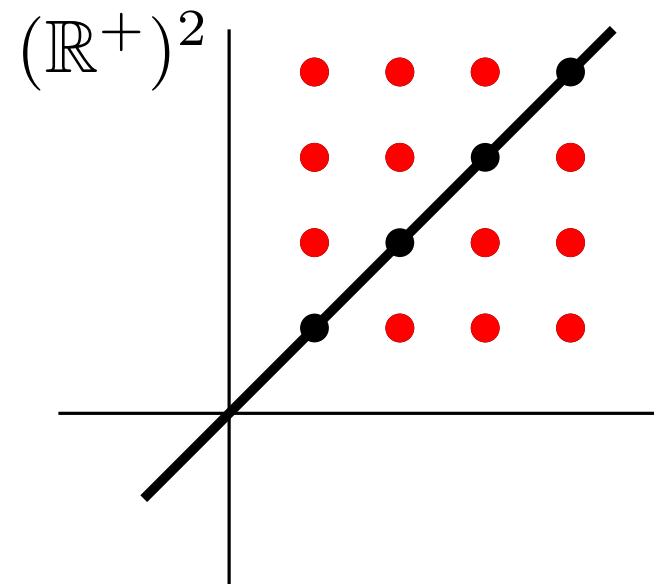
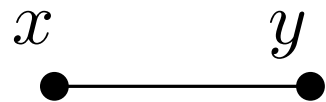
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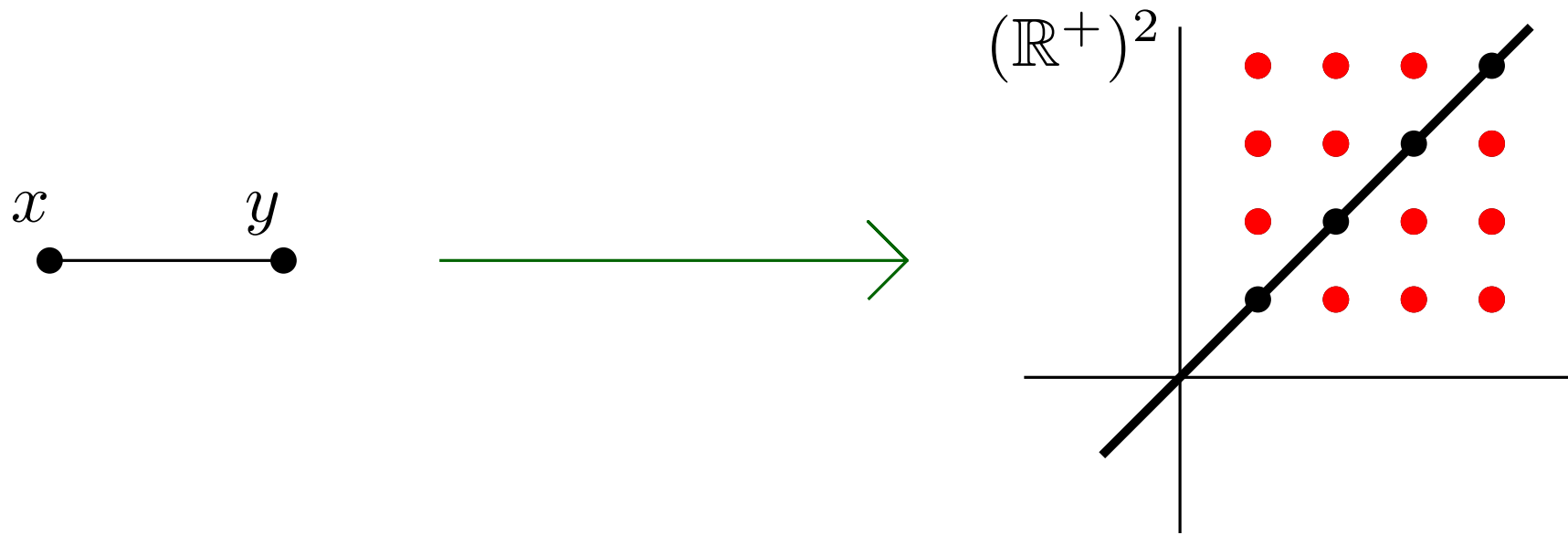


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Call *chambers* the connected components of $(\mathbb{R}^+)^2 \setminus \mathcal{H}_G$

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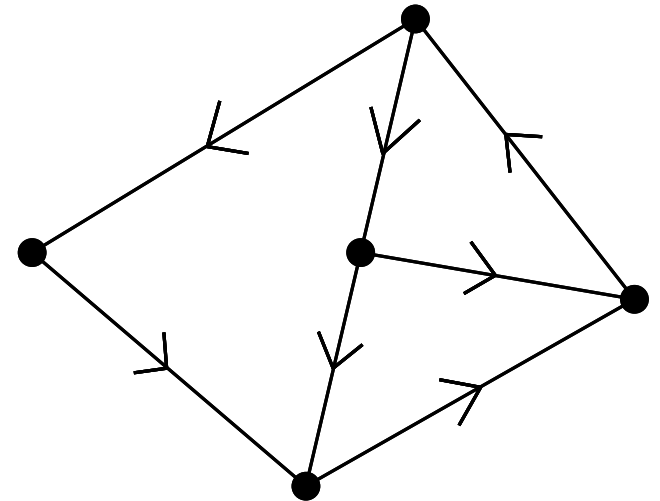


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$$\text{Then } X_G = \sum_{\text{proper } \kappa} x^\kappa = \sum_C \text{ chambers } \sum_{\alpha \in \mathbb{N}^2 \cap C} x^\alpha$$

Ellzey's quasisymmetric invariant

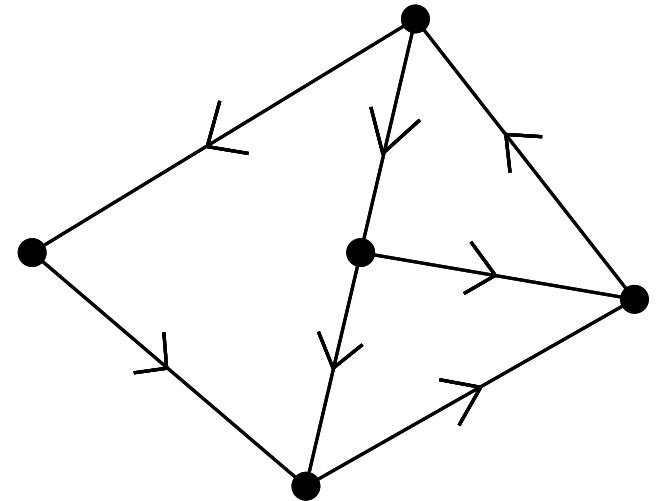
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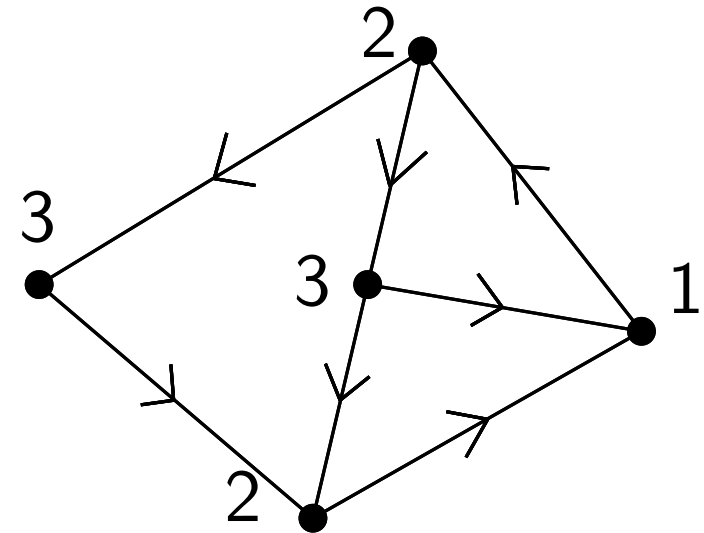
Given a directed graph $\vec{G}$ and a coloring κ of the underlying graph G we define an *ascent* of κ to be a directed edge $(i, j) \in E$ verifying $\kappa(i) < \kappa(j)$



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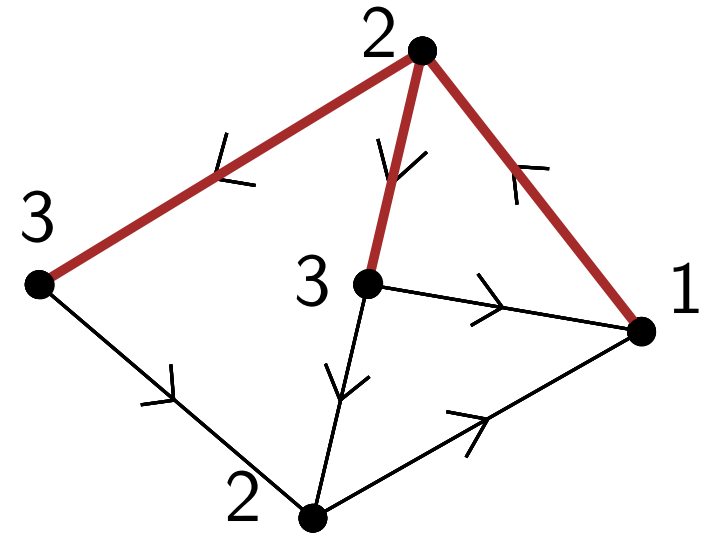
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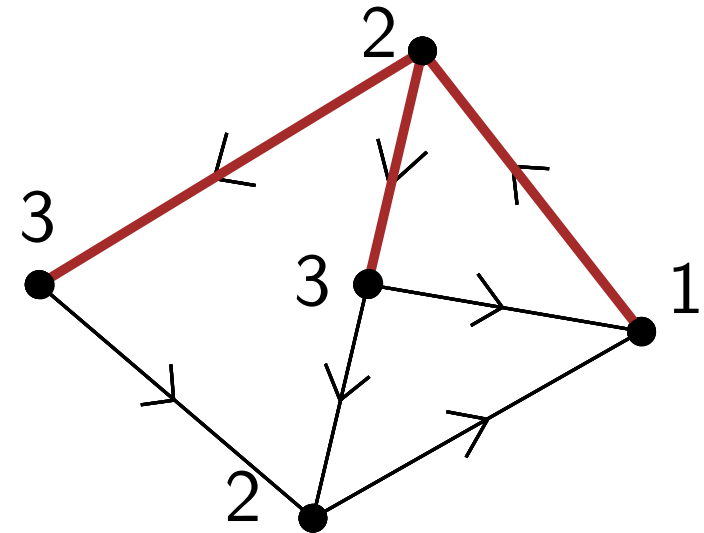


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$$asc(\kappa) = \#\{(i, j) \in E : \kappa(i) < \kappa(j)\}$$

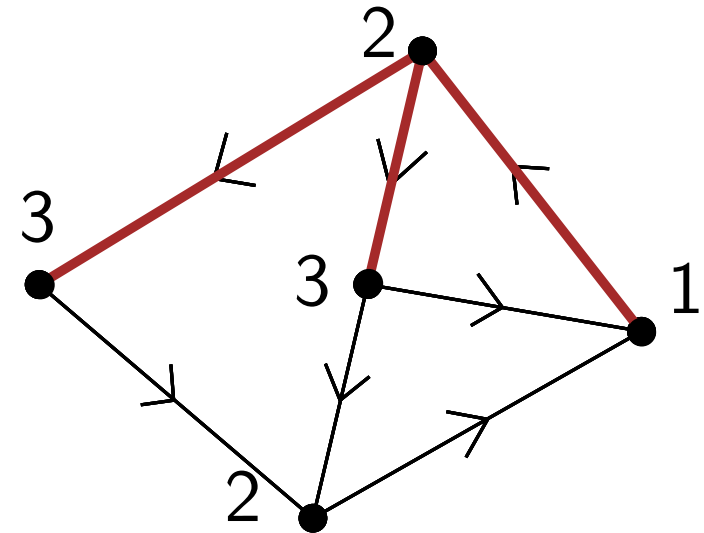


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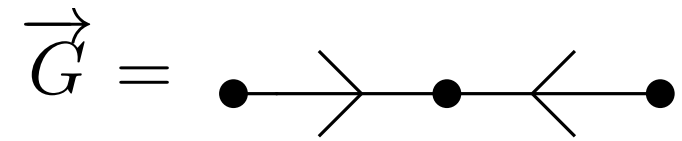


Then the chromatic quasisymmetric function of a digraph is:

$$X_G(x; t) := \sum_{\kappa} x^{\kappa} \cdot t^{asc(\kappa)}$$

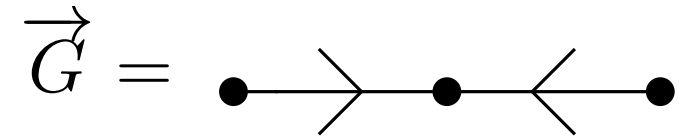
An example

$$X_{\vec{G}} = \sum_{\kappa} x^{\kappa} t^{asc(\kappa)}$$



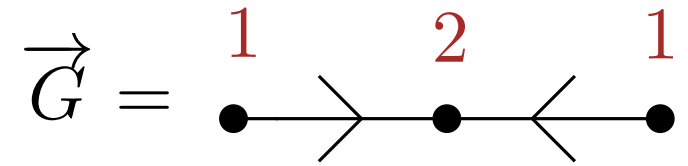
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$$X_{\vec{G}} = \sum_{\kappa} x^{\kappa} t^{\text{asc}(\kappa)} \\ = t^2$$



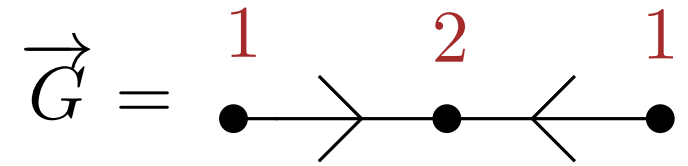
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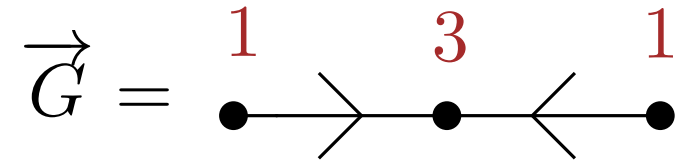
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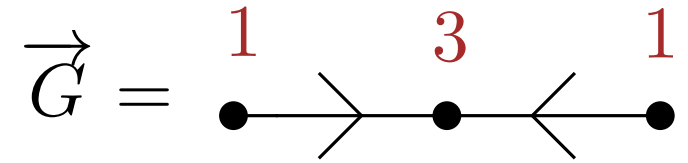
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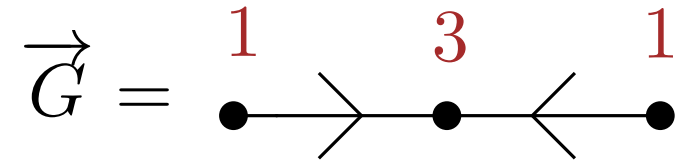
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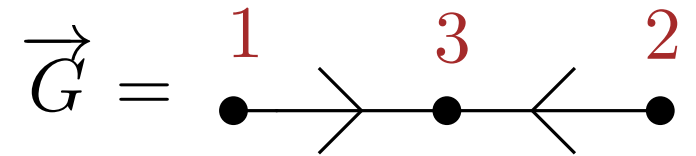
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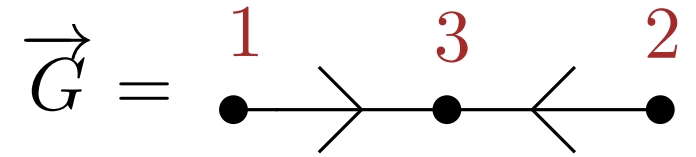
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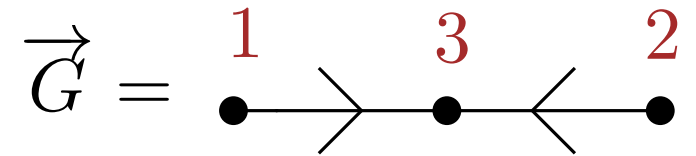
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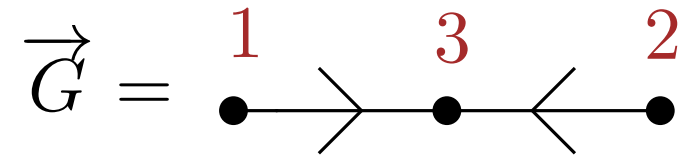
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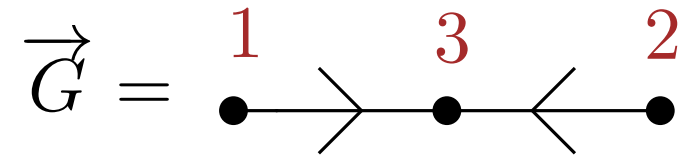


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$$t(2x_1 x_2 x_3 + \cdots) +$$

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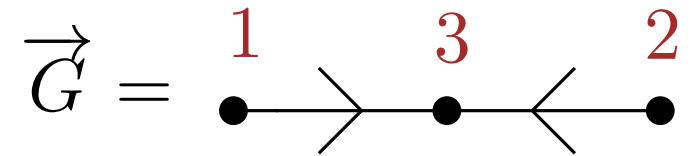
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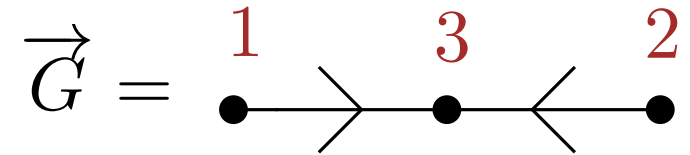
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$$X_{\vec{G}}(x; t) = \sum_{\kappa} x^{\kappa} \cdot t^{asc(\kappa)} \in QSym[t]$$

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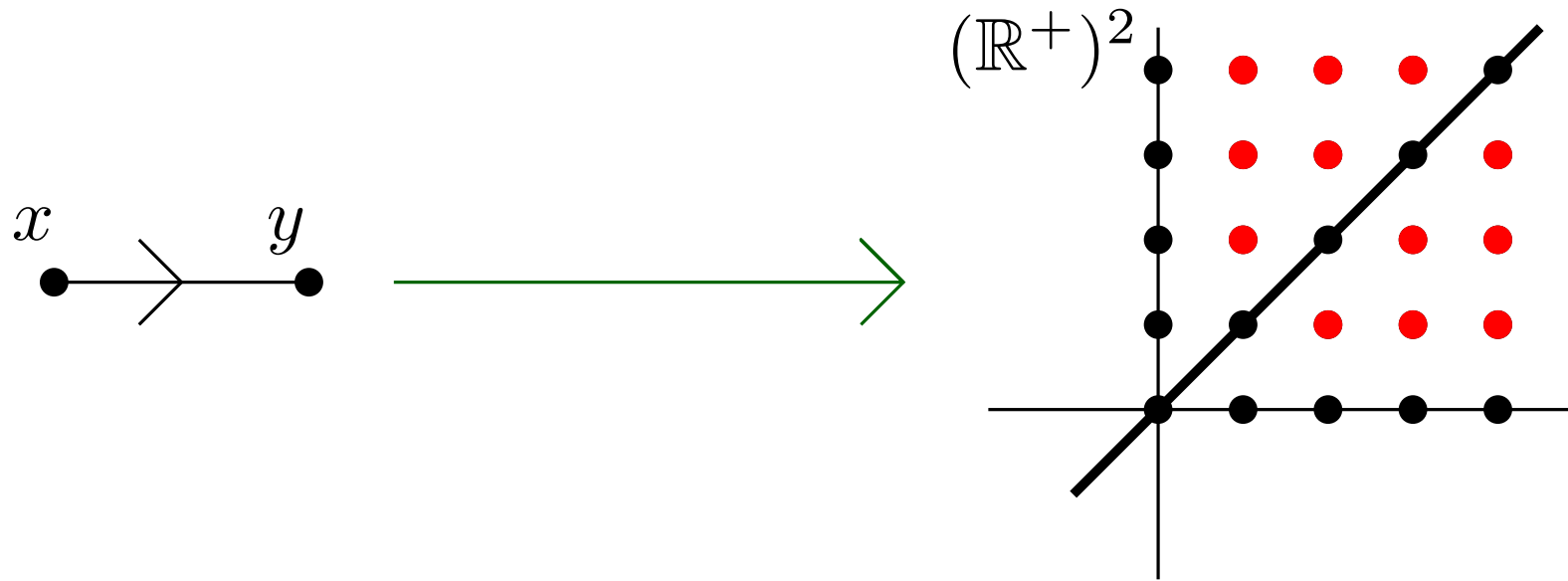
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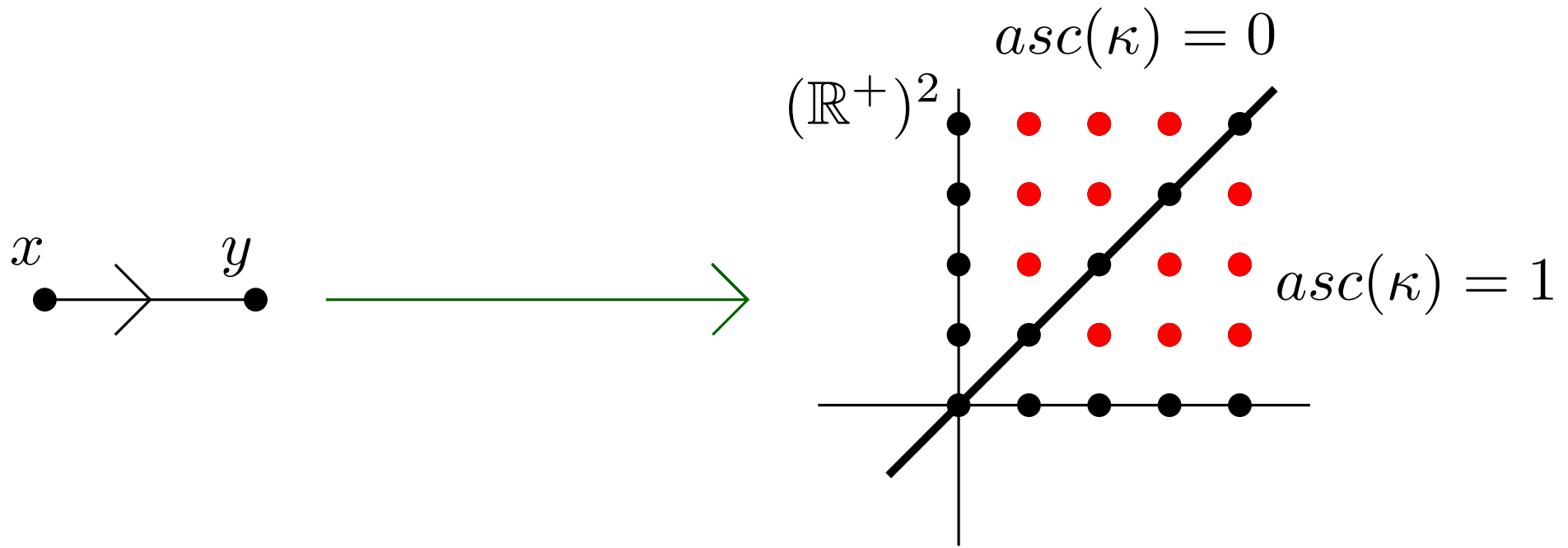
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- The bases of the quasisymmetric function algebra are indexed by compositions $(M_{\alpha}, F_{\alpha} \dots)$
- Quasisymmetric functions are a very important object on algebraic combinatorics

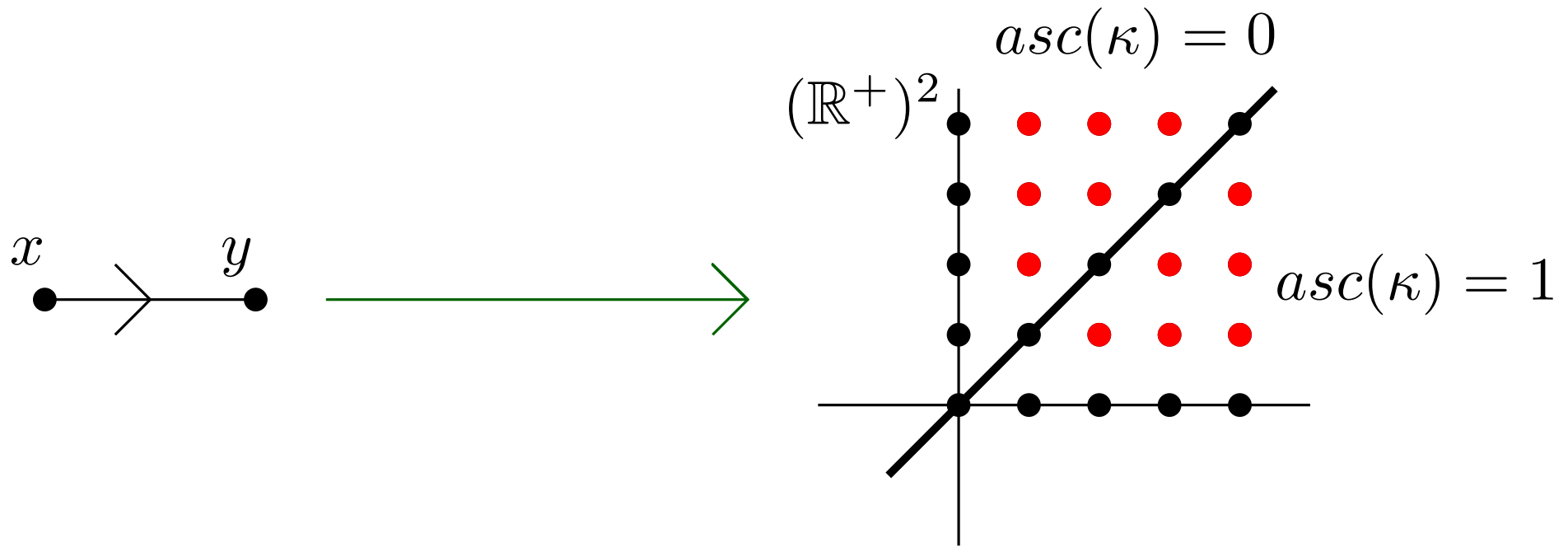
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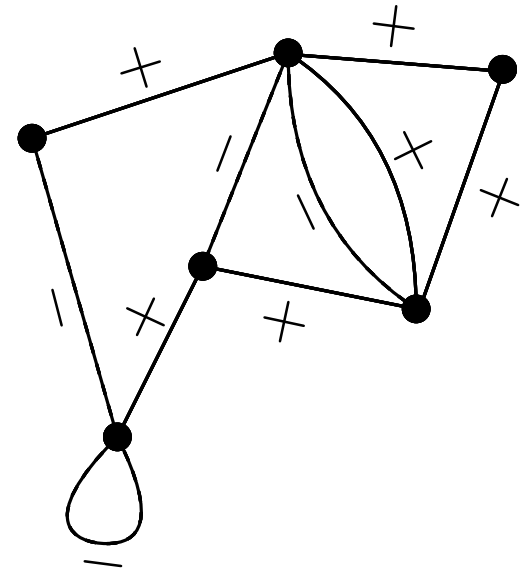


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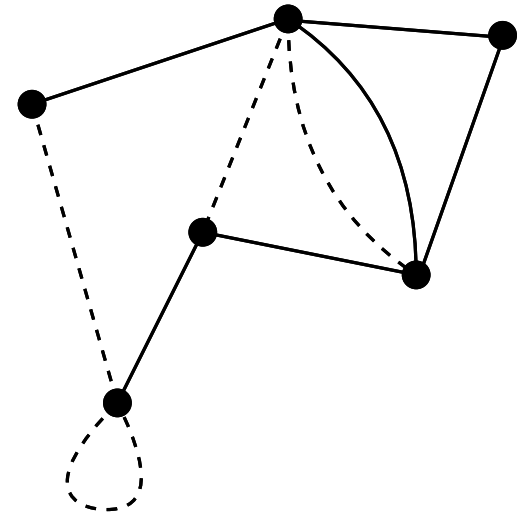


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Signed graphs

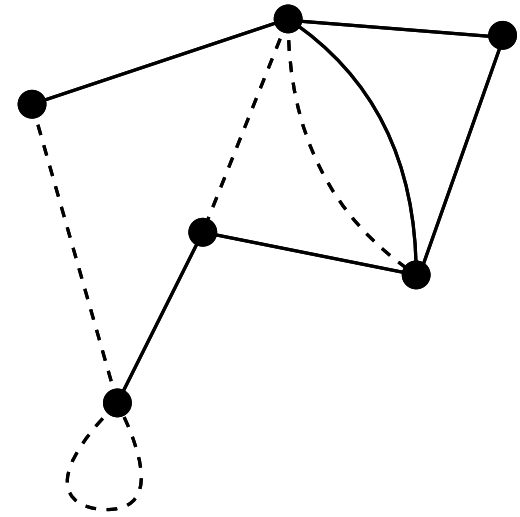


Signed graphs



Signed graphs

What is a proper coloring of a signed graph?



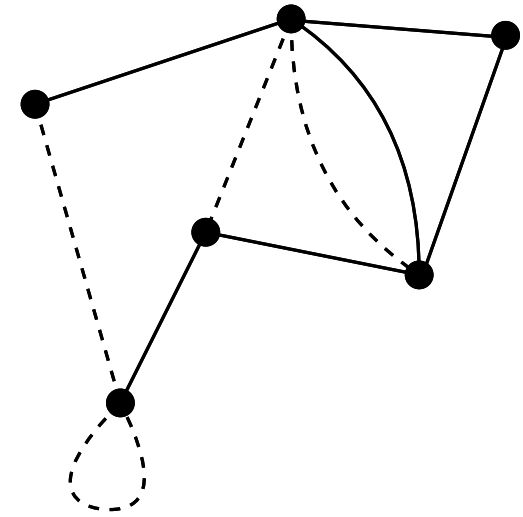
Signed graphs

What is a proper coloring of a signed graph?

A proper coloring of a signed graph

$\Sigma = (V, E^+, E^-)$ is a map $\kappa : V \rightarrow \mathbb{Z}$ verifying:

- $\kappa(i) \neq \kappa(j)$ if $\{i, j\} \in E^+$
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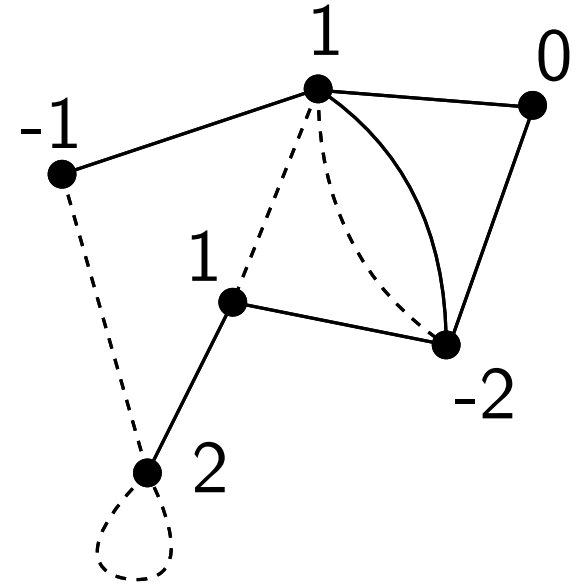
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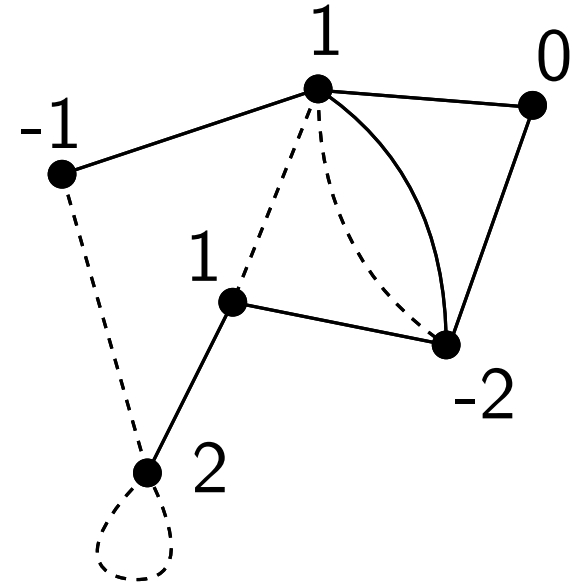


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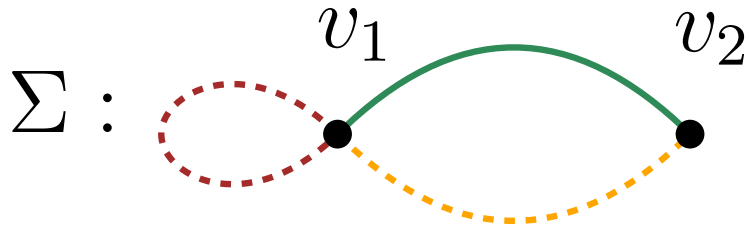
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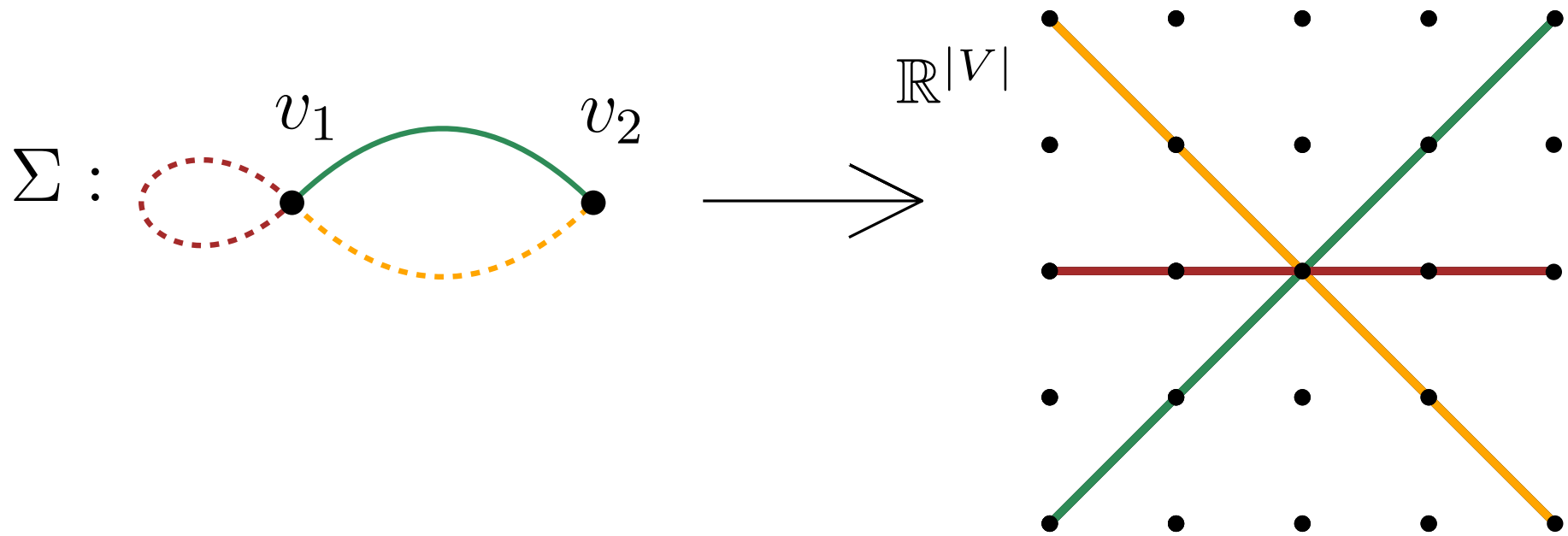
The chromatic polynomial of Σ is

$$\chi_{\Sigma}(k) := \# \left\{ \begin{array}{l} \text{proper colorings of } \Sigma \\ \text{with color set} \\ \{0, \pm 1, \dots, \pm k\} \end{array} \right\}$$

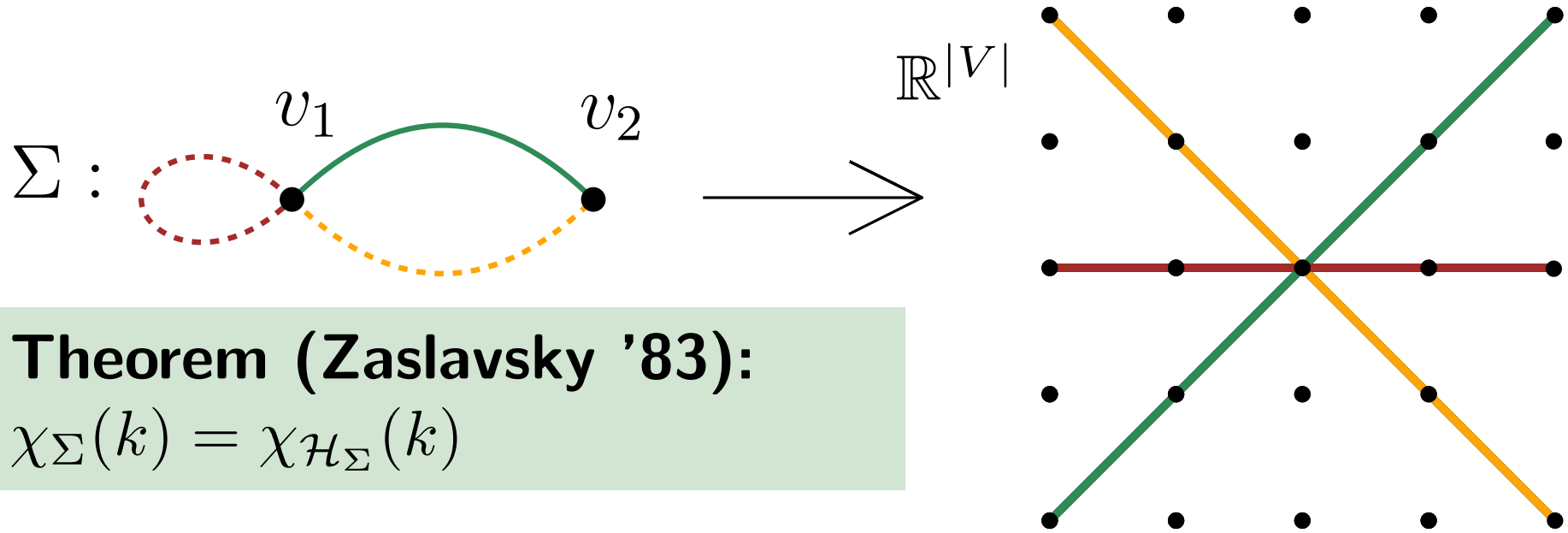
The geometric viewpoint: signed case



The geometric viewpoint: signed case



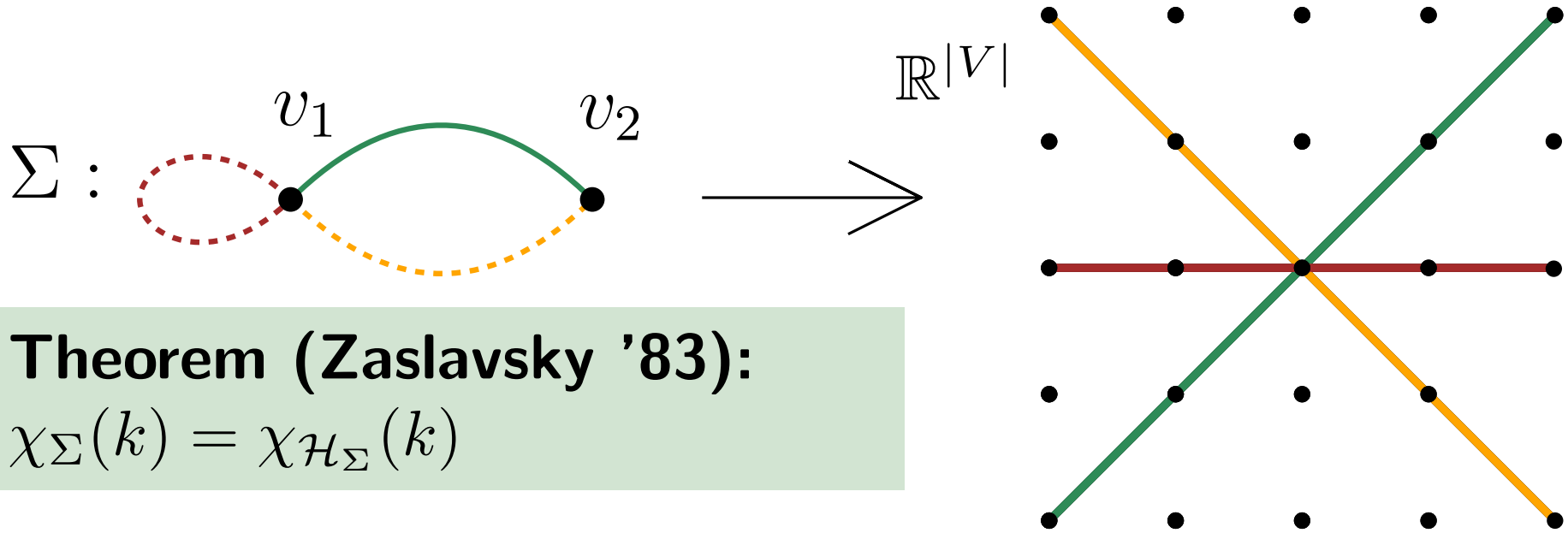
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Theorem (Zaslavsky '83):

$$\chi_{\Sigma}(k) = \chi_{\mathcal{H}_{\Sigma}}(k)$$

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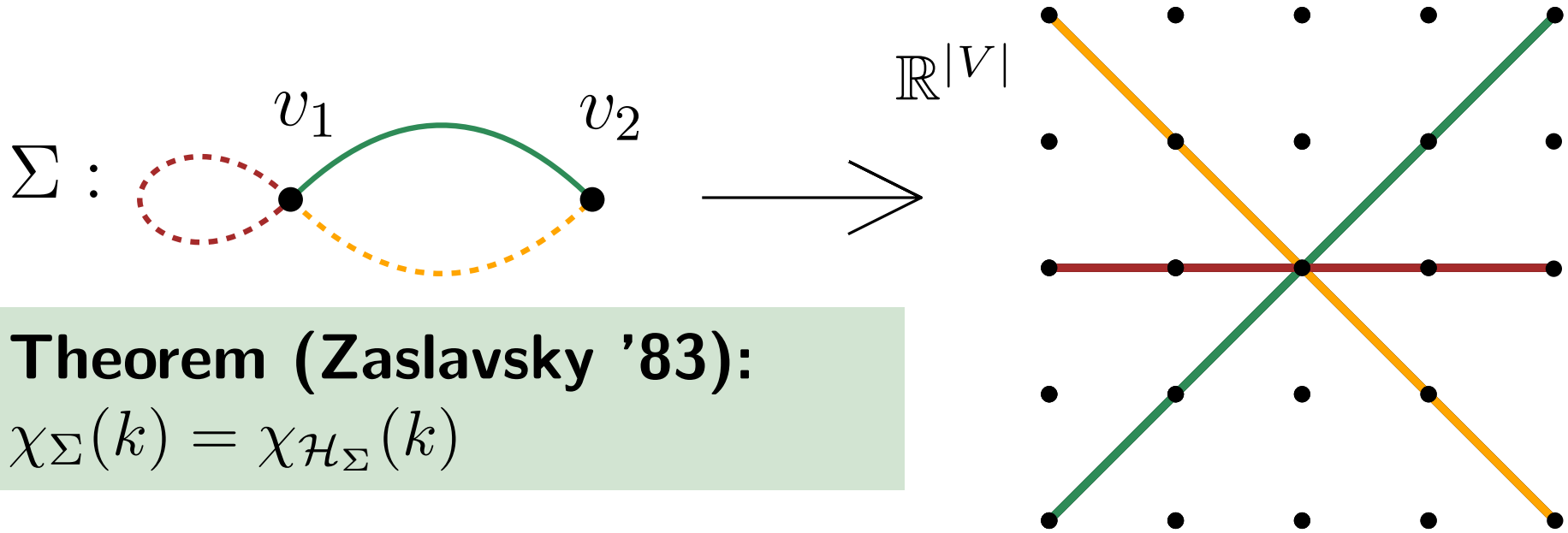
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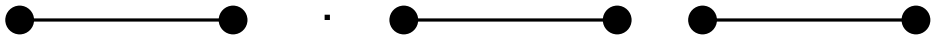
This invariant belongs to $SSym$ the algebra of signed symmetric functions

Orientation of signed graphs

What is an orientation of a signed graph ?

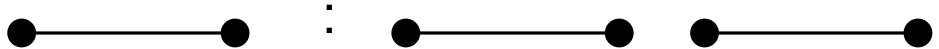
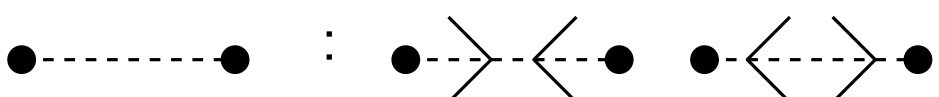
Orientation of signed graphs

What is an orientation of a signed graph ?

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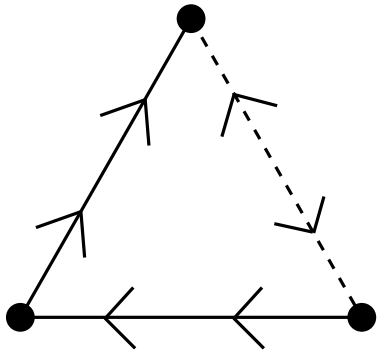
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Orientation of signed graphs

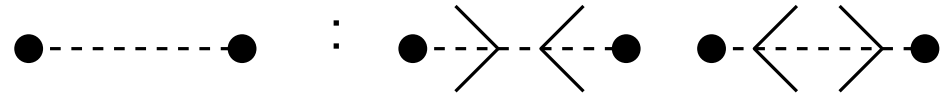
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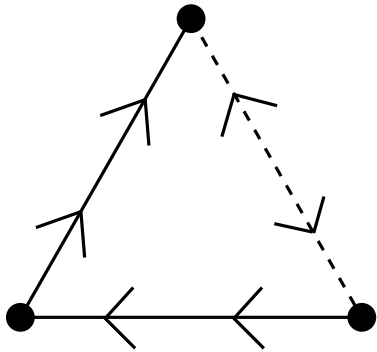


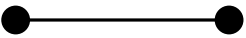
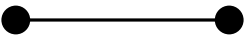
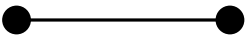
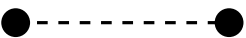
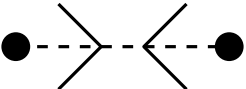
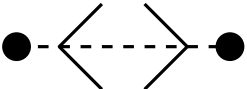
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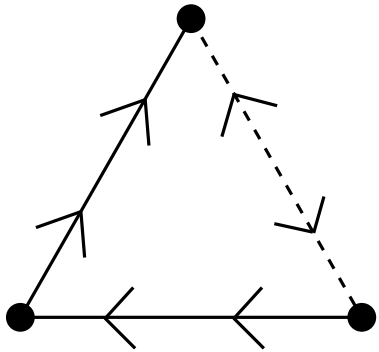


- Positive edges  :  
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An acyclic orientation of Σ is an orientation in which every closed walk has a source or a sink

Orientation of signed graphs

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- Positive edges $\bullet \text{---} \bullet$: $\bullet \rightarrow \bullet$ $\bullet \leftarrow \bullet$
- Negative edges $\bullet \cdots \bullet$: $\bullet \rightarrow \bullet \leftarrow \bullet$ $\bullet \leftarrow \bullet \rightarrow \bullet$

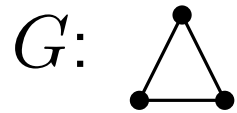
An acyclic orientation of Σ is an orientation in which every closed walk has a source or a sink

Theorem (Zaslavsky):

$$\chi_{\Sigma}(-1) = \#\{\text{acyclic orientations of } \Sigma\}$$

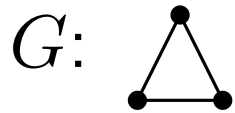
Summary

simple graph



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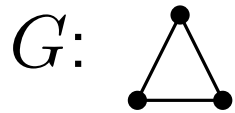


Stanley 1995

$$\longrightarrow X_G(x_1, x_2, \dots) \in Sym$$

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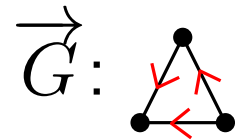
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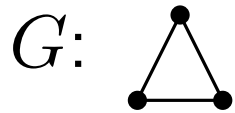


Ellzey 2017

$$\longrightarrow X_{\vec{G}}(x_1, x_2, \dots; \textcolor{red}{t}) \in QSym[t]$$

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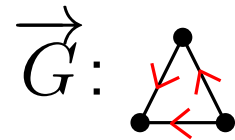
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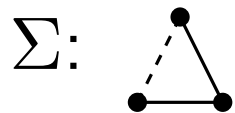
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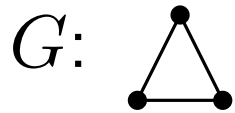
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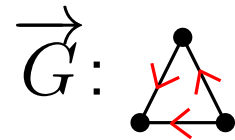
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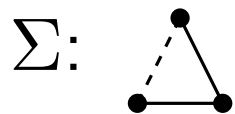
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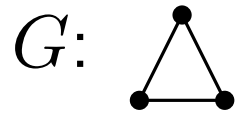


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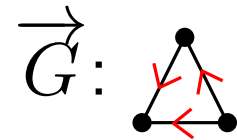
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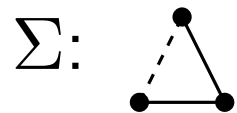
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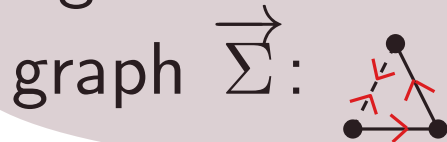
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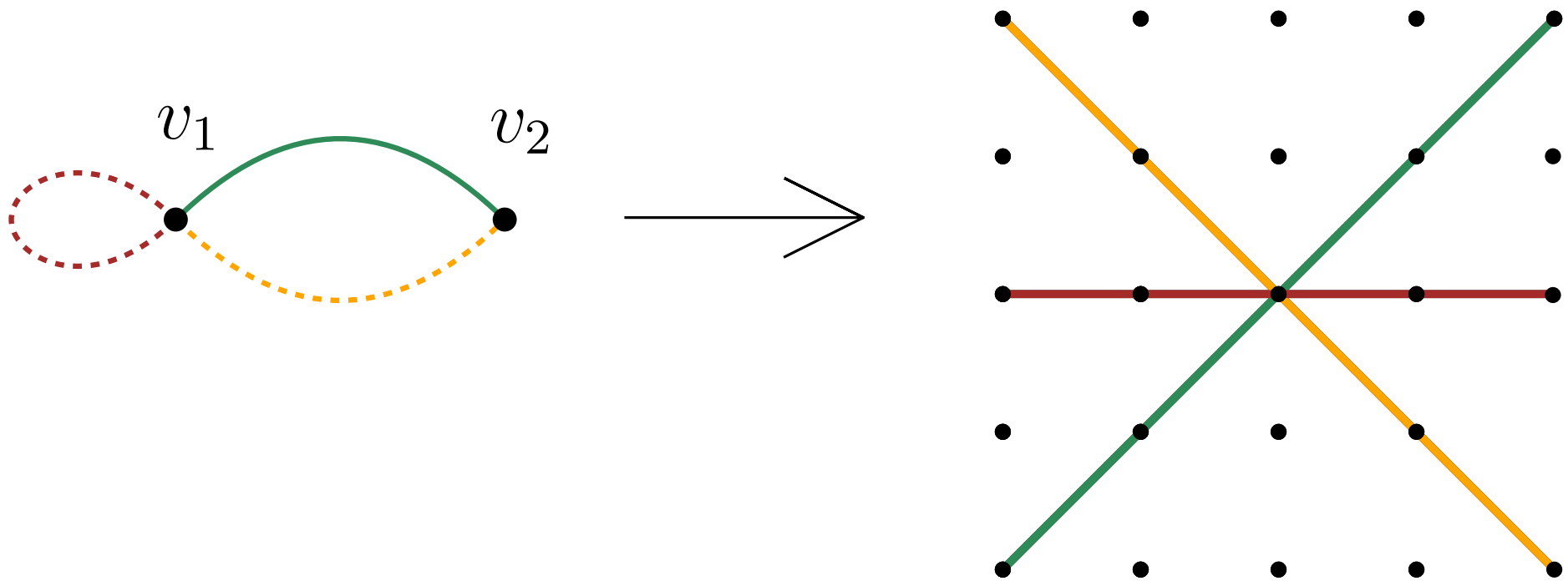
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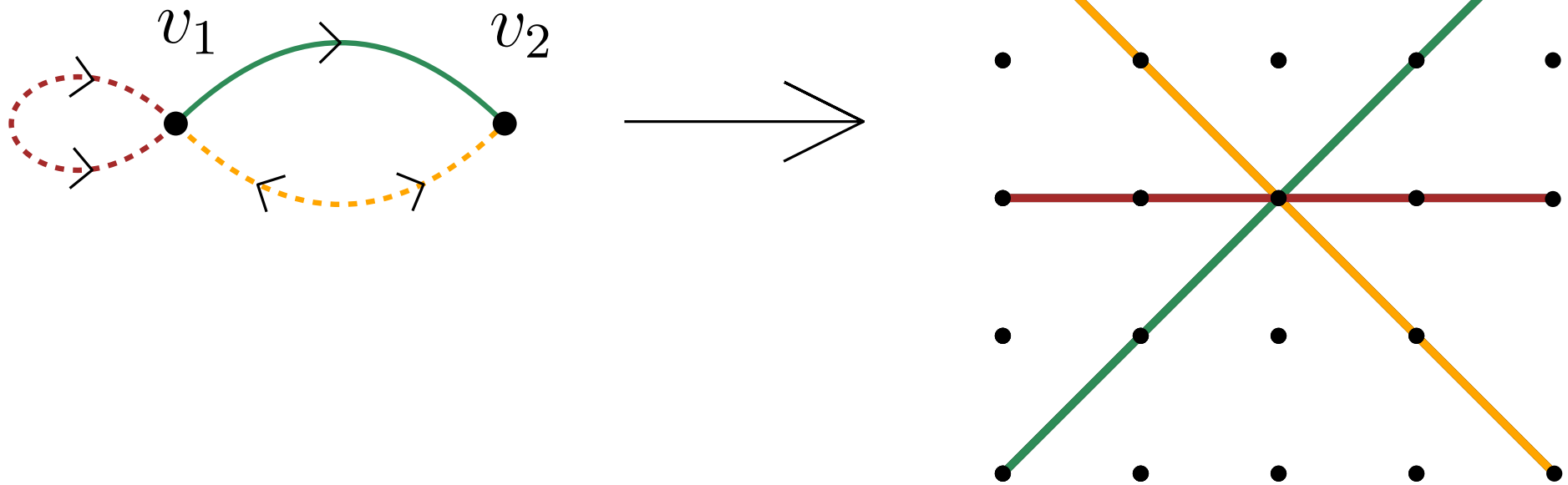


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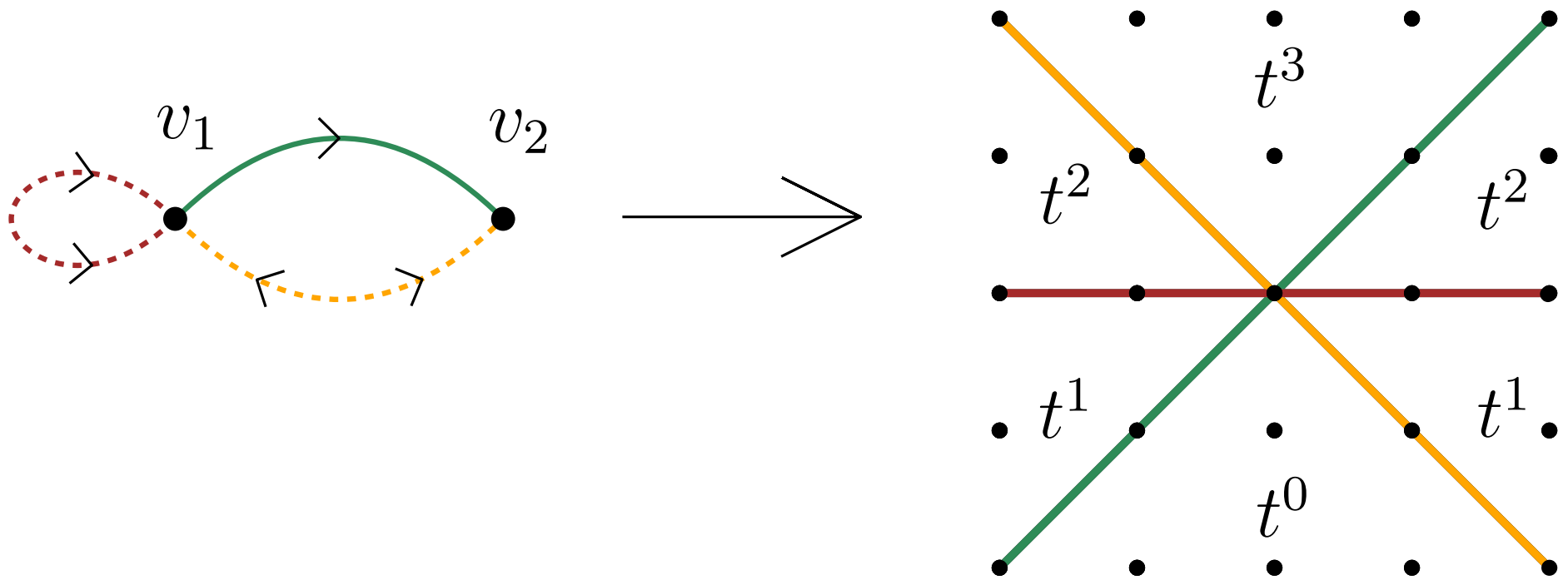
The definition of the invariant



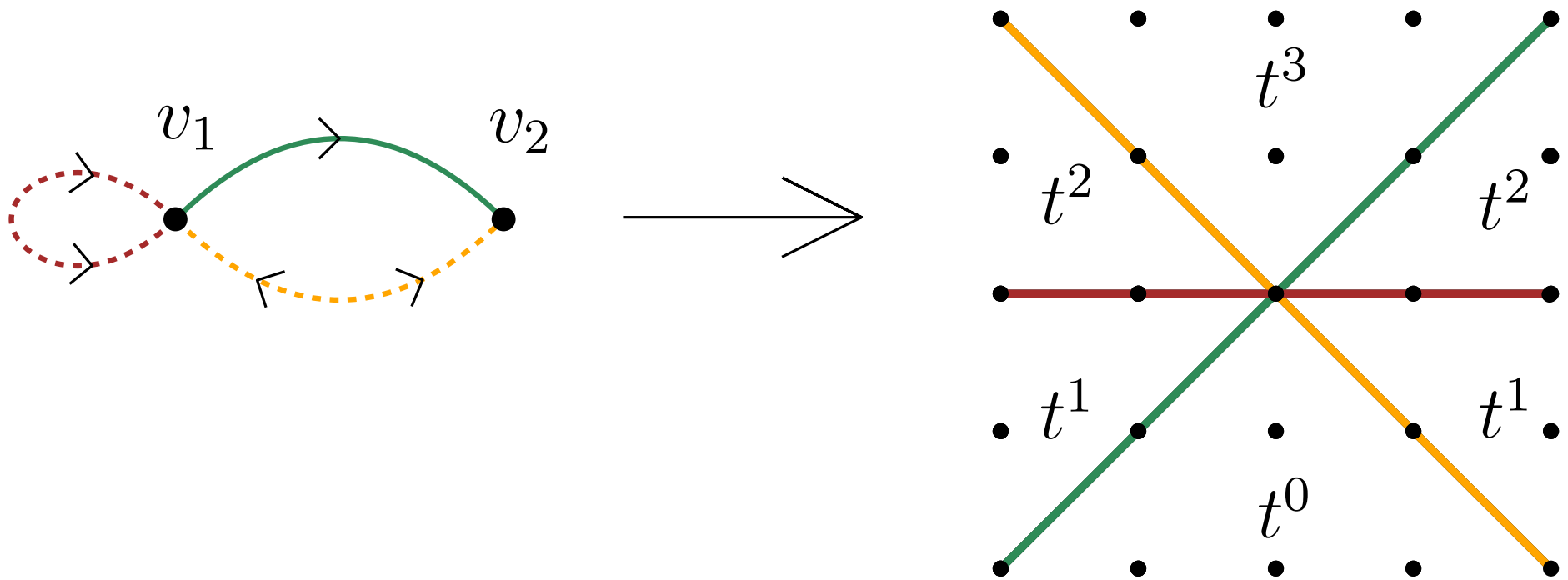
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$$\begin{aligned}
 X_{\vec{\Sigma}}(\mathbf{x}; t) &:= \sum_C t^{\text{asc}(C)} \sum_{\alpha \in \mathbb{Z}^d \cap C} \mathbf{x}^\alpha \\
 &= \sum_{\kappa} t^{\text{asc}(\kappa)} \mathbf{x}^\kappa
 \end{aligned}$$

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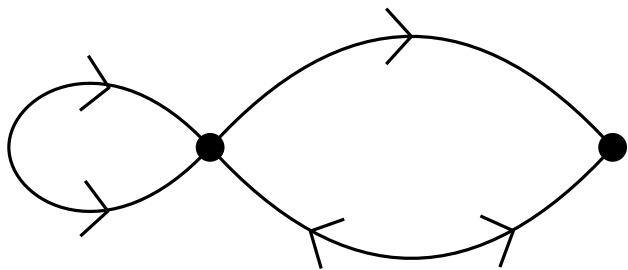
$$\left\{ \begin{array}{c} \text{positive edges} \\ \bullet \xrightarrow{i} \bullet \text{ verifying} \\ \kappa(i) < \kappa(j) \end{array} \right\} \cup \left\{ \begin{array}{c} \text{introverted edges} \\ \bullet \xrightarrow{i} \bullet \xleftarrow{j} \bullet \text{ verifying} \\ \kappa(i) + \kappa(j) < 0 \end{array} \right\} \cup \left\{ \begin{array}{c} \text{extroverted edges} \\ \bullet \xleftarrow{i} \bullet \xrightarrow{j} \bullet \text{ verifying} \\ \kappa(i) + \kappa(j) > 0 \end{array} \right\}$$

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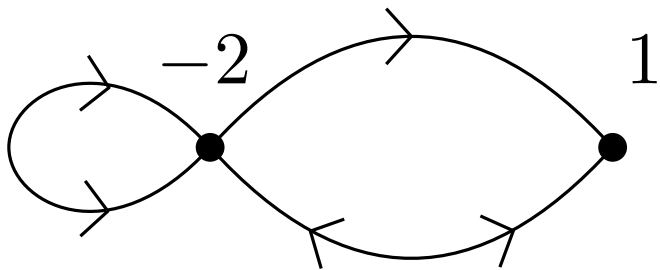


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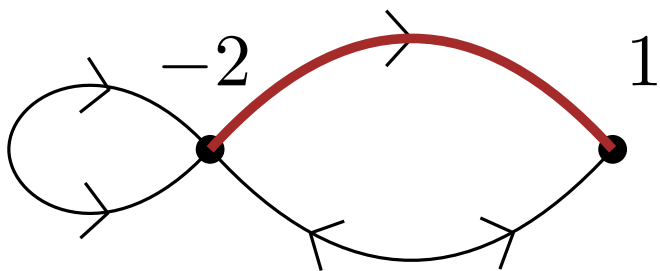


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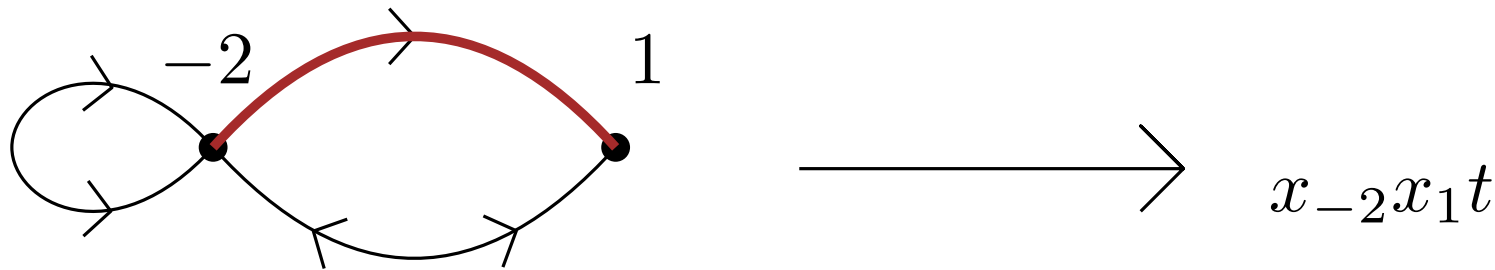


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- Formula for the expansion of $X_{\vec{\Sigma}}$ in this family
- Symmetry question: for which $\vec{\Sigma}$ we have $X_{\vec{\Sigma}} \in SSym$?

The bibliography

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Thank you for your
attention!!

